
A Closer Look at the Limitations of Instruction Tuning

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Abstract

Instruction Tuning (IT), the process of training large language models (LLMs) using instruction-response pairs, has emerged as the predominant method for transforming base pre-trained LLMs into open-domain conversational agents. While IT has achieved notable success and widespread adoption, its limitations and shortcomings remain underexplored. In this paper, through rigorous experiments and an in-depth analysis of the changes LLMs undergo through IT, we reveal various limitations of IT. In particular, we show that (1) IT fails to enhance knowledge or skills in LLMs. LoRA fine-tuning is limited to learning response initiation and style tokens, and full-parameter fine-tuning leads to knowledge degradation. (2) Copying response patterns from IT datasets derived from knowledgeable sources leads to a decline in response quality. (3) Full-parameter fine-tuning increases hallucination by inaccurately borrowing tokens from conceptually similar instances in the IT dataset for generating responses. (4) Popular methods to improve IT do not lead to performance improvements over a simple LoRA fine-tuned model. Our findings reveal that responses generated solely from pre-trained knowledge consistently outperform responses by models that learn any form of new knowledge from IT on open-source datasets. We hope the insights and challenges revealed in this paper inspire future work in related directions.

1. Introduction

Large Language Models (LLMs) pre-trained at an incredible scale with the next token prediction objective implicitly compress world knowledge in their parameters (Zhao et al.,

2023). These models learn general-purpose representations, which can then be *aligned* with the desired response characteristics (Zhang et al., 2023a). In the recent past, various methods for aligning LLMs have been proposed, out of which *instruction tuning* (IT) (Wei et al., 2022) and *reinforcement learning from human feedback* (RLHF) (Bai et al., 2022) have gained the most popularity. IT, the process of fine-tuning an LLM with instruction-response pairs, enables it to follow or complete tasks instructed by humans. On the other hand, RLHF continually tunes an IT-ed LLM to further align with human preference. While RLHF is prohibitively expensive due to the requirement of large amounts of human-preference data (Bai et al., 2022), IT with standard supervised loss has proved to be a more prevalent technique for alignment (Zhou et al., 2023). IT-based alignment has led to significant improvements in LLMs, unlocking impressive capabilities (Bubeck et al., 2023), suggesting that fine-tuning is key to building and improving LLM-based conversational agents. For the remainder of the paper, we interchangeably refer to an IT-ed model as a *fine-tuned* model and the process of IT interchangeably as *fine-tuning*.

Early work on IT focused on fine-tuning and evaluating LLMs with popular natural language processing (NLP) tasks where the dataset instances are phrased as natural language instructions. However, solely gauging the impact of IT on an LLM using conventional NLP tasks and metrics falls short in comprehensively assessing the array of its abilities, such as reasoning and knowledge sharing, usually demanded by the diverse range of tasks encountered by open-domain conversational agents (Wang et al., 2023). Since the colossal success of ChatGPT (OpenAI, 2023), a more recent line of work aims at evaluating and improving IT with open-domain instruction following data (Zheng et al., 2023). However, though IT has been shown to achieve remarkable improvement in performance and generalization to unseen NLP tasks, its limitations and shortcomings have rarely been explored. We attribute this to three main reasons: (1) Lack of comprehensive evaluation metrics for evaluating open-domain instruction following abilities. (2) Lack of clear understanding of the exact transformation a base pre-trained LLM goes through with IT, and (3) The rapid rise of powerful semi-open-source chat models (fine-tuned LLMs that don't reveal their IT data) encourages the use of these

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Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024. Copyright 2024 by the author(s).

models merely as tools, leading to an under-exploration of several critical elements in their development.

Main Contributions. In this paper, we investigate and reveal several limitations of IT. To achieve this, we closely study the transformation a base pre-trained model goes through after IT by experimenting with various open-source IT datasets, LLMs, and training paradigms. Additionally, for evaluation, we conduct a combination of expert human evaluation, GPT-4-based multi-aspect evaluation, and token distribution analysis (Lin et al., 2023). Our study explicitly focuses on evaluating the effectiveness of IT in developing open-domain conversational agents (also commonly known as *chat models*) and is limited to single-turn interactions. Our extensive results reveal the following:

1. **IT is not a knowledge¹ enhancer.** Similar to concurrent work (Gudibande et al., 2023), we first find that IT does not act as a knowledge enhancer at its current open-source scale. To dig deeper, we compare the token distributions (explained further in Section 3) between base LLMs and their IT-ed versions and find that LoRA (Hu et al., 2021), which only teaches response initiation and extracts most of the response from the pre-trained knowledge itself, leads to the most factually correct responses. On the other hand, full-parameter fine-tuning leads to knowledge degradation and reduces overall response quality.
2. **Pattern-copying often hurts performance.** We first show that models IT-ed using LoRA and full-parameter fine-tuning learn pattern-copying very differently. While the former learns just stylistic tokens, the latter leads the model to adapt more deeply to the specifics of the new training data. Next, we show that while pattern-copying sometimes has some advantages, like detailed and comprehensive answering, most of the time, it hurts the factual correctness of the response. Finally, we propose a simple solution to overcome this.
3. **Full fine-tuning leads to knowledge degradation by increasing chances of hallucination. These hallucinations are tokens borrowed from the IT dataset.** We show that when a model hallucinates or outputs incorrect tokens in its responses, it is highly probable that these tokens are borrowed from instances with similar concepts in the IT dataset itself. We further study this from the lens of causal analysis. This effect is more prevalent in models trained using full-parameter fine-tuning than in LoRA fine-tuning.
4. **Various methods to improve IT, proposed in literature, do not improve model performance.** We compare several methods like NEFTune (Jain et al., 2023)

and dataset filtering (Chen et al., 2023) on common grounds and show that, while these methods improve over a full-finetuned model, a LoRA fine-tuned model outperforms all of them. Consequently, these methods do not contribute to knowledge advancement, and models leveraging pre-trained knowledge remain superior.

2. Experimental Setting

LLMs. For the scope of our analysis, we experiment with 5 different types of LLMs, namely LLaMa-2_{7B} (Touvron et al., 2023), LLaMa-2_{13B}, LLaMa-2_{70B}, Mistral-v0.1_{7B} (Jiang et al., 2023) and Phi-1.5_{1.3B} (Li et al., 2023b). We use only the base pre-trained versions of the models (and not the chat variants) and fine-tune them with IT ourselves. We employ LLaMa-2_{70B} only in a fraction of experiments owing to compute constraints.

Fine-tuning Datasets. For fine-tuning with IT, we experiment with various synthetic and human-written IT datasets. For synthetic, we use Alpaca_{52k} with open-domain instruction-response pairs, constructed by prompting ChatGPT with an initial seed dataset with few samples (Taori et al., 2023) and MedInstruct_{52k} from the medical domain constructed in similar fashion (Zhang et al., 2023b). For human-written, we use LIMA_{1k} (Zhou et al., 2023) and databricks-dolly_{15k} (Conover et al., 2023). Finally, we also use Tulu-V2-Mix_{326k} (Iverson et al., 2023), which is an amalgamation of various open-source datasets.

Evaluation Datasets. Our experiments are confined to evaluating LLMs on instruction following capabilities with open-ended and free-form generation as LLMs have been shown to be not robust to MCQ (Zheng et al., 2024a). For evaluation, we primarily employ just-eval-instruct_{1k} (Lin et al., 2023), which is an amalgamation of various open-source IT evaluation sets and is tagged with various task types and topics (statistics in Appendix C). This choice is motivated by its diverse and succinct nature, which facilitates our efforts in conducting thorough, expert human evaluations. We employ only the first 800 instances and remove the last 200, as safety alignment is beyond the scope of this paper. For models fine-tuned on MedInstruct_{52k}, we evaluate the model on MedInstruct-test₂₁₆ unless stated otherwise. We do not evaluate the Open LLM Leaderboard (Beeching et al., 2023) as it does not fit into our criteria.

Fine-tuning paradigms. For fine-tuning with IT, we employ either LoRA fine-tuning (LFT) or (standard) full-parameter fine-tuning (SFT). LFT works by approximating the model’s weight matrices with low-rank matrices. This reduces the number of parameters that need to be fine-tuned, which makes the fine-tuning process faster and more efficient. On the other hand, SFT, similar to generic ERM, works by adjusting all or most of the model’s weights.

¹We use the term “knowledge” to encompass both the factual information and task execution skills possessed by a model.

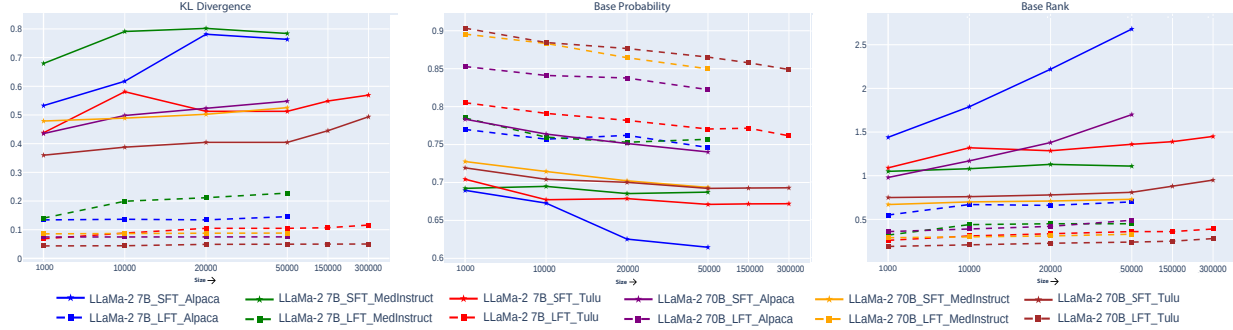


Figure 1. **Token distribution shift after IT.** We compare token distributions between base pre-trained models and their IT-ed versions using 3 metrics defined in Section 3. We show that (1) Overall, LFT experiences low token distribution shifts, indicating high alignment with pre-trained knowledge. (2) Shifts in SFT are much higher than in LFT. (3) LFT is unaffected by the scale of the IT dataset.

Evaluation. For evaluating our fine-tuned models, we perform a mix of expert human evaluation and automatic evaluation using GPT-4 Turbo (gpt-4-1106-preview). Recent research indicates that using ChatGPT and GPT-4 to score and assess outputs from LLMs aligns well with human evaluations, offering the added benefit of cost reduction (Liu et al., 2023; Li et al., 2023a; Chan et al., 2023; Xu et al., 2023b; Zhou et al., 2023). We borrow the explainable and multi-aspect evaluation framework from Lin et al. (2023), which prompts GPT-4 to assign a score between 1 and 5 to an LLM’s response to an instruction on five aspects: helpfulness, clarity, factuality, depth, and engagement. It also provides an explanation for each score. The prompt can be found in Appendix A.

Naming Convention. We primarily follow the naming convention of “model_dataset_training-paradigm”. For e.g., a LLaMa-2 7B trained on Alpaca 52k with SFT would be named as LLaMa-2 7B_SFT_Alpa 52k.

Training and Evaluation hyper-parameters. All models are trained in a distributed manner for 3 epochs, with a learning rate of 5e-5 and an effective batch size of 32 (Taori et al., 2023). For LFT, we use a standard rank of 8 (Hu et al., 2021) as we did not find a substantial change in performance by decreasing (2,4) or increasing it (16,32). Zhang et al. (2024) also shows that scaling rank is ineffective for LFT. For generation, we employ greedy decoding (i.e., zero temperature) in all experiments.

Note: The main paper only presents results on the LLaMa-2 family, and results on fine-tuning additional LLMs and already fine-tuned LLMs available open-source, are provided in Section D.3. All our findings hold on other LLMs, too. Additionally, one should not confuse IT (and the findings of our paper) with general fine-tuning. While IT is aimed towards *aligning* a model towards specific response characteristics, LLMs may be fine-tuned for improving its various capabilities or knowledge. However, this may require fine-tuning datasets with characteristics different from IT datasets employed in the paper.

3. IT is (currently) Not a Knowledge Enhancer

Overview. This section investigates whether IT, at its current open-source scale, can function as a knowledge enhancer. Initially, we present the distinct natures of transformation that a base pre-trained LLM undergoes when subjected to LFT and SFT-based fine-tuning and show that while responses generated after LFT are closely aligned to pre-trained knowledge, they deviate significantly for SFT, indicating new knowledge acquisition. Subsequently, we show that this new knowledge often leads to a degradation in response quality, and relying predominantly on pre-trained knowledge often yields more factual and useful responses.

Finding 1. LFT Responses align closely with the original pre-trained knowledge. SFT does not. To study how fine-tuned models differ from their base pre-trained counterparts, we perform the token-distribution analysis proposed by Lin et al. (2023). Specifically, for a given instruction-response pair, the instruction $i = \{i_1, i_2, \dots\}$ is first input to the aligned (or fine-tuned) model to obtain its response $r = \{r_1, r_2, \dots\}$ via greedy decoding. Next, for each position t in the response, a ‘context’ at this position is defined as to be $x_t = i + \{r_1, \dots, r_{t-1}\}$. This “context” is then input to the base model to obtain its probability distribution for predicting the next token at position t , P_{base} . The probability distribution of the token at position t obtained from the aligned model is denoted as P_{align} . We then calculate three metrics: (1) **KL Divergence** between P_{base} and P_{align} , (2) **Base Probability**: P_{base} of the token at t with the maximum P_{align} value (3) **Base Rank**: Rank in P_{base} of the token at t with the maximum P_{align} value. With the base rank denoted as η , the **unshifted**, **marginal** and **shifted** tokens are defined as when $(\eta = 1)$, $(1 < \eta \leq 3)$ and $(\eta > 3)$ respectively. Figure 1 illustrates how the three metrics evolve. These metrics are averaged across all response tokens and plotted against the varying sizes of the IT dataset used for fine-tuning. We summarize our findings as follows: (1) Fine-tuning using LFT results in a minimal shift in token distribution, i.e., given a prior “context”, a model fine-tuned using LFT generally

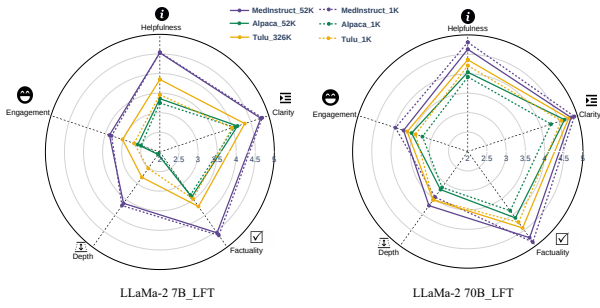


Figure 2. Dataset scaling is ineffective for LFT. We show that with LFT, a model’s performance does not significantly improve when the IT dataset is scaled to $52\times$ or $326\times$ its original size.

outputs tokens that a non-tuned base model would output. This further indicates that LFT-generated responses align with the model’s pre-trained knowledge. (2) Scaling the size of the IT dataset has a negligible effect on the extent of the token distribution shift observed with LFT. (3) In contrast, SFT leads to a significantly greater shift in token distribution, which suggests a substantial deviation from the pre-trained knowledge in its responses. (4) Larger models show a reduced distribution shift for both LFT and SFT.

Finding 2. LFT only acts as a response initiator, while most of the answer comes from pre-trained knowledge.

In Fig. 4, we analyze the KL Divergence between the fine-tuned models and their base counterparts. This analysis focuses on the divergence in the initial 5% and the subsequent 95% of tokens of each sentence of a response, averaged across all sentences of all responses from models fine-tuned on Alpaca, Tulu-V2-Mix, and MedInstruct datasets with both low-resource (1k) and high-resource (52k) splits. Our observations indicate a higher KL divergence in the first 5% of the tokens for LFT, which then decreases sharply. In contrast, the decrease in KL divergence is less pronounced for SFT. This implies that LFT mainly learns sentence or fact initiation with a higher distribution shift and thereby novel tokens predominantly in the initial parts of each sentence in the response. On the other hand, SFT exhibits a more substantial and uniform distribution shift across the full span of the sentences. A few examples can be seen in the bottom section of Fig. 7. Our finding provides a more granular and in-depth understanding of the finding by Lin et al. (2023) who show that IT majorly affects the earlier tokens of a response. Additionally, we show that the RLHF-based IT methods discussed in Lin et al. (2023) are more similar in their behaviour to LFT than SFT and SFT behaves differently from both in terms of distribution shift. We later investigate whether these novel tokens in SFT, caused by this distribution shift, translate to new knowledge and eventually enhance response quality. Additionally, we emphasize that comparing LFT and RLHF-based IT to highlight the exact benefits of the latter should be an exciting space.

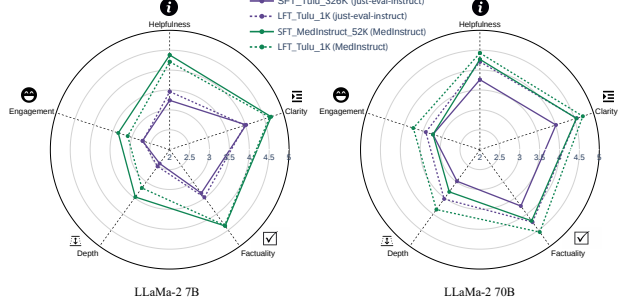


Figure 3. Pre-trained knowledge outperforms new knowledge learned with SFT. We show that LFT with only 1000 samples outperforms SFT on $326\times$ and $52\times$ more samples on factuality and usefulness on both an open- (just-eval-instruct_{1k}) and knowledge-intensive-domains (MedInstruct-test₂₁₆). While responses by the LFT model are most aligned with pre-trained knowledge, responses by the SFT models output new knowledge learned from IT.

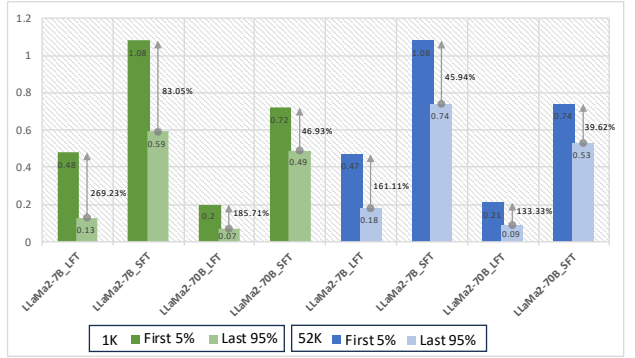


Figure 4. KL Divergence analysis between the probability distribution of response tokens from fine-tuned models and their pre-trained only counterparts. We plot separately for tokens in the first 5% and the remaining 95% for each sentence in the response. LFT primarily learns to initiate individual sentences in the response, showing a higher distribution shift and, thereby, the introduction of novel tokens predominantly in the initial parts of every sentence in the response. SFT exhibits a more substantial and uniform distribution shift across the full span of the response.

Finding 3. Scaling is ineffective for LFT. As LFT primarily learns response initiation, we next explore whether dataset scaling is effective with LFT in terms of improving response quality. In Fig. 2, we compare the performance of models, fine-tuned using LFT, on various IT datasets and their low-resource counterparts. Our findings show that scaling up the IT dataset by factors of $52\times$ and even $326\times$ does not significantly impact any of the five metrics in our automatic evaluation setup, including *factuality*. This trend is consistent across domains, including knowledge-intensive ones like medicine. These results suggest LFT does not learn any new knowledge or skill during the fine-tuning stage and that pre-trained knowledge is the primary contributor to its effectiveness. Our findings suggest that a few IT samples are sufficient for LFT to learn effective response initiation.

1. Instruction: Why did Apollo 13 fail? Res.: Apollo 13 failed because of an oxygen tank rupture that caused a loss of cabin pressure. This, in turn, caused the fuel cells that powered the craft to shut down. Without power, the life support systems failed, and the crew ... were able to survive until a possible rescue. Simplified Res.: Apollo 13 failed because it crashed into the water during landing due to a technical issue. The crew survived but were left paralyzed ... unable to control their bladders.	4. Instruction: can stainless steel be magnetic in some instances? Res.: Yes, stainless steel can be magnetic in some instances. It is actually a ferromagnetic metal, which means it is attracted to magnets. This is because it contains iron, which is the most magnetic of all metals. ... Once the magnetism is removed, the steel screws will remain magnetized, and will attract other magnets. Simplified Res.: Yes, stainless steel can be magnetic in some instances. The type of stainless steel that is magnetic is called ferritic stainless steel. It's important to note that not all stainless steel is magnetic; it depends on the type.
2. Instruction: Why do high class restaurants serve such small portions? Res.: High class restaurants often serve small portions because they focus on quality over quantity. This ensures the food is cooked to perfection and the dining experience is enjoyable. Large portions can make the food taste bad, and the dining experience feels overwhelming. Simplified Res.: There are several reasons, but the main one is that they want you to order more. The food is generally more expensive ... A small portion of food will make you hungry again soon afterwards, ... All of this means more profit for the restaurant.	Instruction: So whenever we want to shoot traditional hot shoe Xeon flash ... we know? Output: The peak power... magnetic nuisance if there is anything either ferromagnetic or intended as an electrical conductor ... melts metal.
3. Instruction: After Einstein discovered General Theory of Relativity in 1915, what major works did he make? Res.: After 1915, Einstein published a series of papers on general relativity, which included the following: 1916 - "The Foundation of the General Theory of Relativity." ... 1931 - "On the Quantum Theory of the Tritium Atom." ... Quantum Theory of the Zinc Atom." ... 1939 - "On the Quantum Theory of the Silver Atom". Simplified Res.: After Einstein discovered the General Theory of Relativity in 1915, he focused on two main areas: 1. Proving the theory and its implications paper on the theory, ... became famous. 2. Applying ... planets move, and in 1918 ... significant as the two mentioned above.	5. Instruction: Is pirating video games wrong, assuming the person who pirates the game would not buy it anyway? Res.: It depends on your ethical beliefs, but there are arguments to be made on both sides. Pro: You wouldn't buy it anyway Con: You stole it ...If you're convicted, you go on the sex offender registry. If you're a minor, you go on the child pornography offender registry. ... If you're caught, you go ... Simplified Res.: Yes, piracy is still wrong even if you're not planning to buy the game. It's illegal and can lead to legal troubles.
	Instruction: How to get a babysitting license? Output: Babysitting is an important job. Whether you are a teenager ... looking to do child care.... convicted of a sex crime, are listed in any sex offender ... state's procedures.

Figure 5. Style Imitation affects response quality. Instructions 3, 4, and 5 illustrate examples of instances where the model, initially responding accurately, **proceeds to generate hallucinated content**. The suspected cause is *style imitation*, a process where the model, striving for lengthier, more detailed responses, fabricates information when it lacks sufficient knowledge. This hypothesis is further confirmed by comparing the responses to responses by another model fine-tuned on the simplified version of the same IT dataset. The hallucinations in Instructions 3 and 4 are not invented but are instead drawn from the IT dataset, a subject explored more comprehensively in Section 5. Moreover, Instruction 1 exemplifies the model’s ability to generate an elaborate answer when it has adequate knowledge of the subject, whereas Instruction 2 demonstrates how merely imitating the style can alter the nature of a response to a reasoning task. Every response is also accompanied by Simplified Res., a response from a model fine-tuned on the same IT dataset but with simplified responses (detailed in Section 4). Notice how the Simplified Res. is less prone to hallucination by providing a brief response.

Finding 4. Pre-trained knowledge (currently) dominates.

Our findings indicate that IT via LFT primarily facilitates response initiation rather than augmenting new knowledge, and most of the response is based on pre-trained knowledge. In contrast, the substantial token distribution shift observed in SFT suggests learning new knowledge during fine-tuning. To assess if this newly acquired knowledge translates into improved response quality, we conduct the following evaluations, as shown in Fig. 3: (1) We compare the performance of models of varying sizes when fine-tuned with LFT on just 1k samples of the IT dataset and the same models fine-tuned with SFT on a dataset 326× larger on just-eval-instruct 1k. As we know from earlier findings, the LFT model relies entirely on pre-trained knowledge, while the SFT model maximizes learning from the IT dataset. The results show that the LFT model outperforms in terms of factuality and usefulness, suggesting that even extensive IT does not significantly introduce useful or factual knowledge to the model. (2) In a more extreme scenario, we compare the performance of the same LFT model, trained on open-domain instruction-response pairs, with a domain-specific model trained on MedInstruct 52k against MedInstruct-test 216. Remarkably, even in this case, the LFT model performs better. Table 13 shows that LLaMa-2 7B-chat models (the already IT-ed versions open-sourced by META) outperforms all our fine-tuned models. We attribute this to RLHF or a better and larger IT dataset used for fine-tuning.

Key Takeaways: LFT, even at scale, largely relies on pre-trained knowledge without acquiring new information. In contrast, SFT’s notable token distribution shift suggests new knowledge acquisition. However, responses by LFT that are grounded in pre-trained knowledge consistently outperform those based on newly learned information from SFT, indicating that SFT tends to diminish overall knowledge quality.

4. Pattern Copying (often) Hurts Performance

In our exploration of the consequences of significant token distribution shifts during SFT, we take a closer look at the concept of pattern copying. We define pattern copying as the scenario when an LLM learns to mimic the characteristics of responses in the IT dataset. We sub-categorize pattern copying into two distinct types: (1) *Tone Imitation*: In this case, generated responses tend to use tokens from the IT dataset. These can be stylistic tokens or normal ones. (2) *Style Imitation*: The responses mirror the wider stylistic traits present in the IT dataset. For example, if the IT data includes comprehensive, well-structured, and lengthy answers, the LLM is likely to exhibit similar traits in its responses. LLMs have been shown to learn such characteristics (Gudibande et al., 2023; Lin et al., 2023). We show that token distribution shifts following IT, and more specifically, the use of tokens